Software Testing: A Research Travelogue (2000–2014)

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Partially supported by: NSF, IBM, Google, and MSR

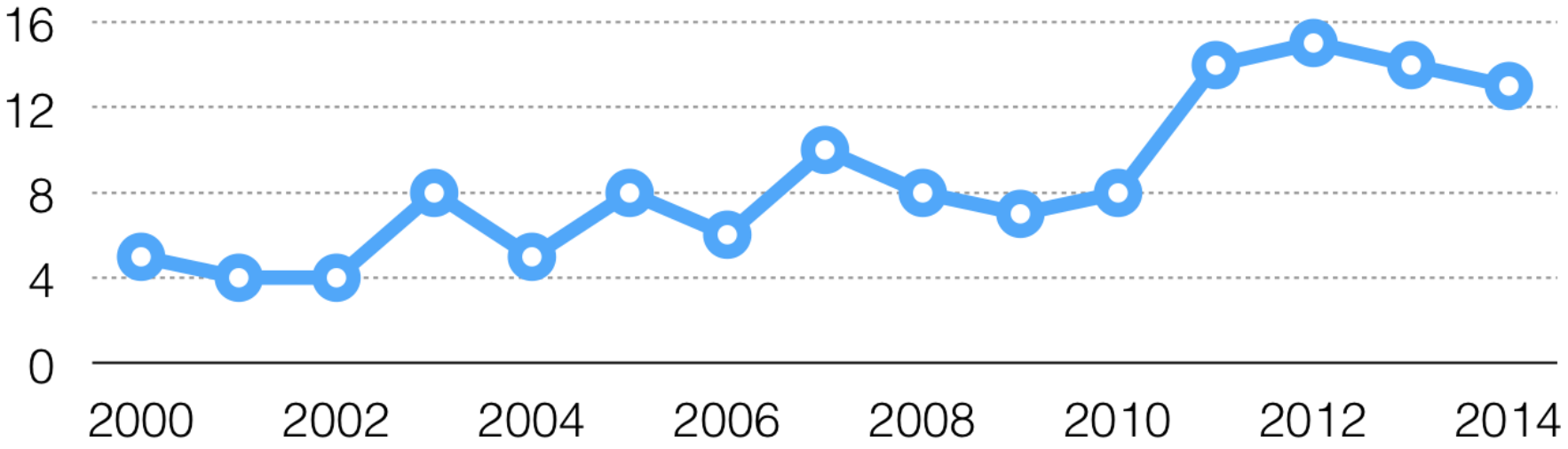


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Software Testing

and investigated

Most used approach

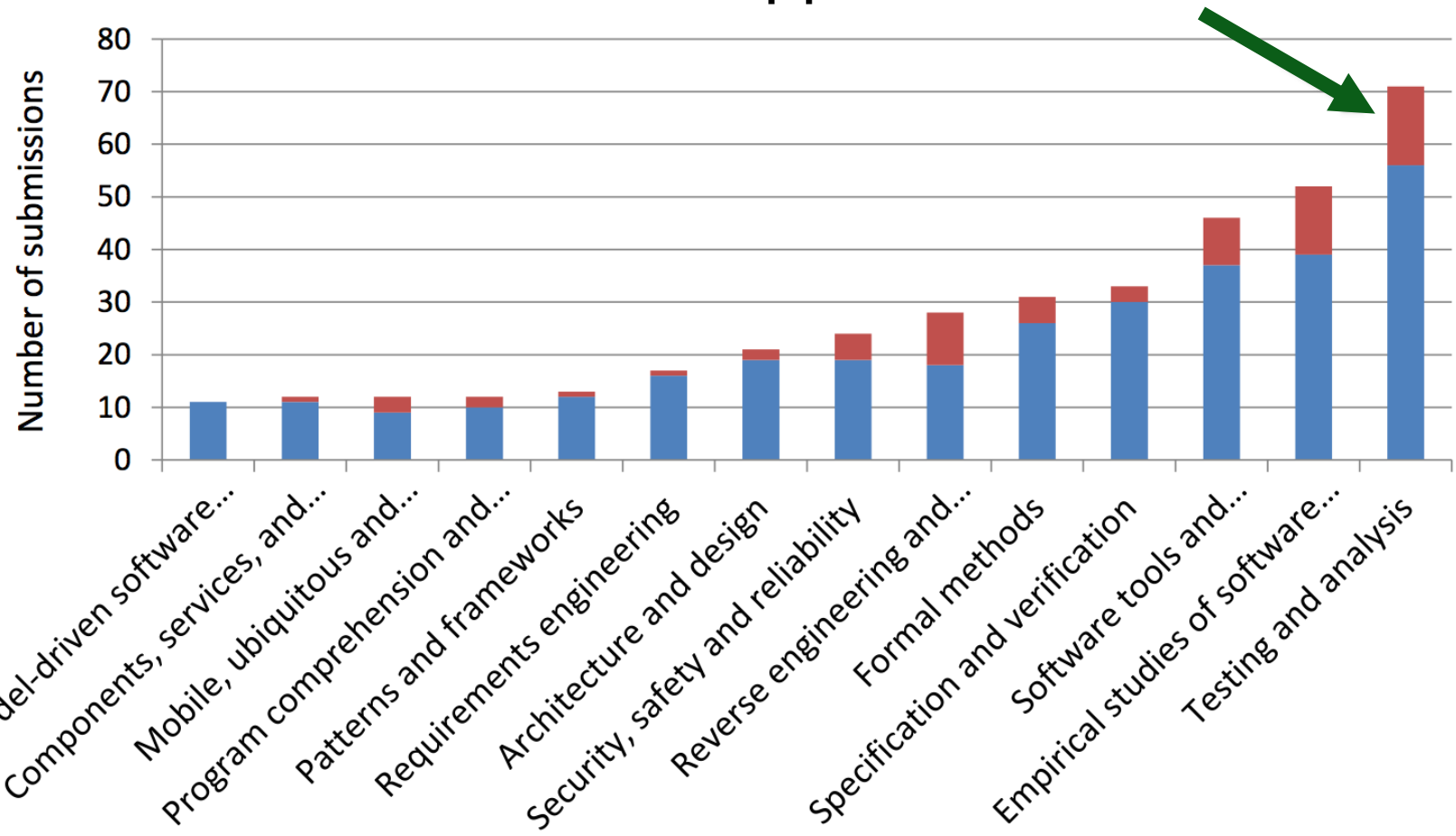


Number of papers on testing at ICSE from 2000 to 2014

Software Testing

and investigated

Most used approach



Testing – A Travelogue

A Special Acknowledgement

Mary Jean Harrold



(1947–2013)

A Travelogue – Our goals



Discuss most successful research in software testing since 2000



Identify most significant challenges and opportunities

A Travelogue – Our Approach



Two questions

1. What do you think are the most significant contributions to testing since 2000?
2. What do you think are the biggest open challenges and opportunities in this area?

In a Nutshell



Research Contributions

Automated Test Input Generation

- Dynamic Symbolic Execution
- Search-based Testing
- Random Testing
- Combined Techniques

Testing Strategies

- Combinatorial Testing
- Model-Based Testing
- Mining/Learning from Field Data

Empirical Studies & Infrastructure

Regression Testing

Practical Contributions

- Frameworks for Test Execution
- Continuous Integration



Challenges & Opportunities

- Testing Real-World Systems
- Oracles
- Probabilistic Approaches
- Testing Non-Functional Prop.
- Domain-Based Testing
- Leveraging Cloud and Crowd

So Many Things, So Little Time…

Automated Test Input Generation

Regression Testing

Empirical Studies & Infrastructure

Practical Contributions

Leveraging Cloud and Crowd



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Automated Test Input Generation

Input



{ Achieve coverage goal
Reach a given point/state

Not new, but resurgence

- Symbolic execution
- Search-based testing
- Random/fuzz testing
- Combined techniques

Technical improvements
Powerful machines
Powerful decision procedures
Careful engineering

Automated Test Input Generation



{ Achieve coverage goal
Reach a given point/state

Not new, but resurgence

- Symbolic execution
- Search-based testing
- Random/fuzz testing
- Combined techniques

Technical improvements
Powerful machines
Powerful decision procedures
Careful engineering

Symbolic Execution

symbolic state:
$x=x_0$, $y=y_0$
path condition:
true

```
1. int testme(int x, int y) {
2.    int z;
3.    if (x < 0)
4.       return -x;
5.    z = x - y;
6.    if (y > z)
7.       fail();
8.    return z;
9. }
```

symbolic state:
$x=x_0$, $y=y_0$, $z=x_0-y_0$
path condition:
$x_0 \geq 0$, $y_0 > x_0/2$



solution:
$x_0=1$, $y_0=1$

Symbolic Execution

symbolic state:
$x=x_0$, $y=y_0$
path condition:
true



```
1. int testme(int x, int y) {
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Symbolic Execution

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true

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1. int testme(int x, int y) {
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5.    z = cmplxFnct(x, y);
6.    if (y > z)
7.       fail();
8.    return z;
9. }
```

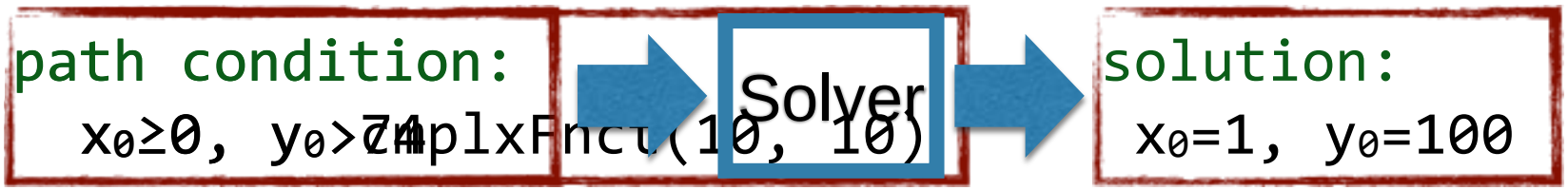
path condition:
$x_0 \geq 0$, $y_0 > \text{cmplxFnct}(10, 10)$



Dynamic Symbolic Execution

Open Challenges

- Highly structured inputs
- External libraries
- Large complex programs
- Oracle problem



So Many Things, So Little Time…

Automated Test Input Generation

Regression Testing

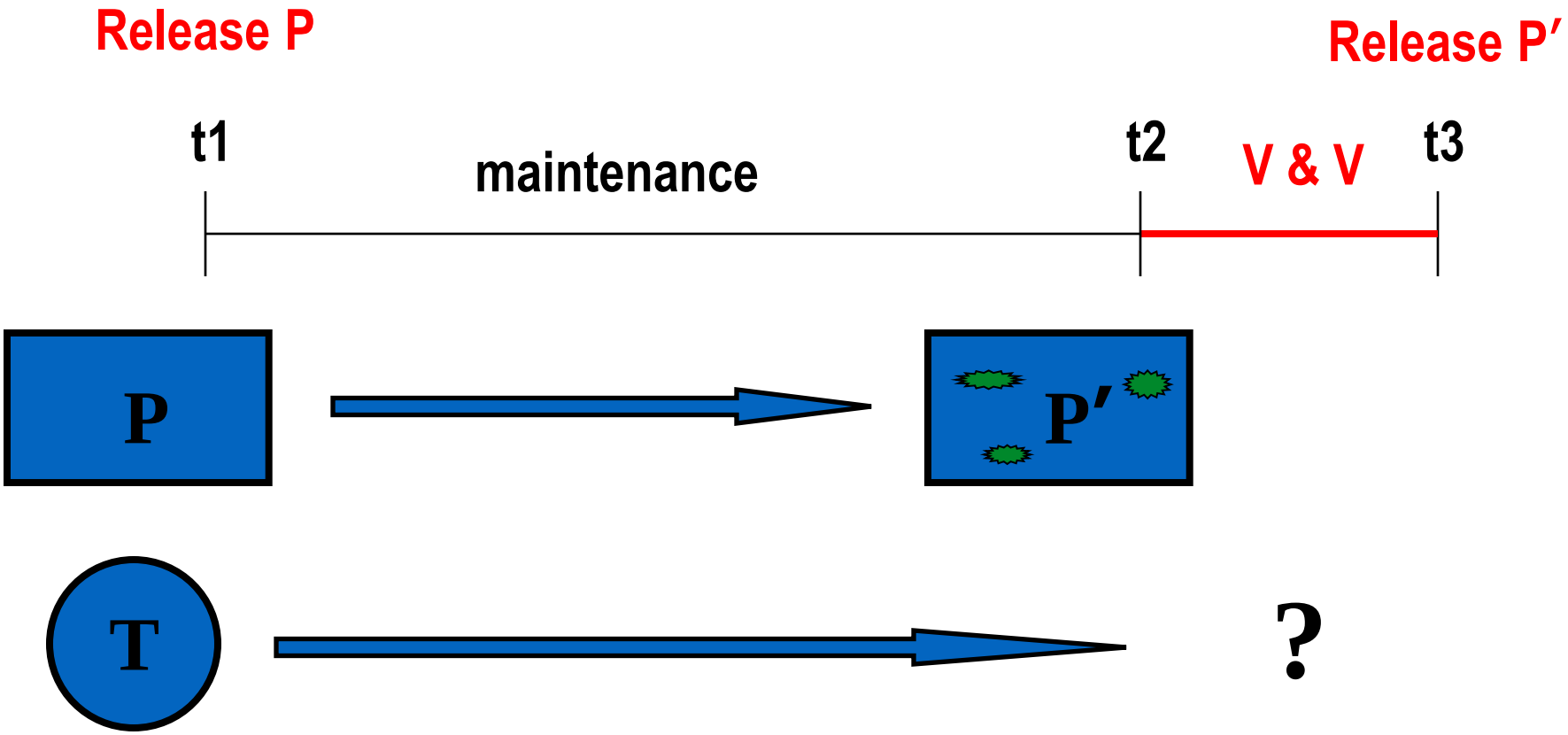
Empirical Studies & Infrastructure

Practical Contributions

Leveraging Cloud and Crowd



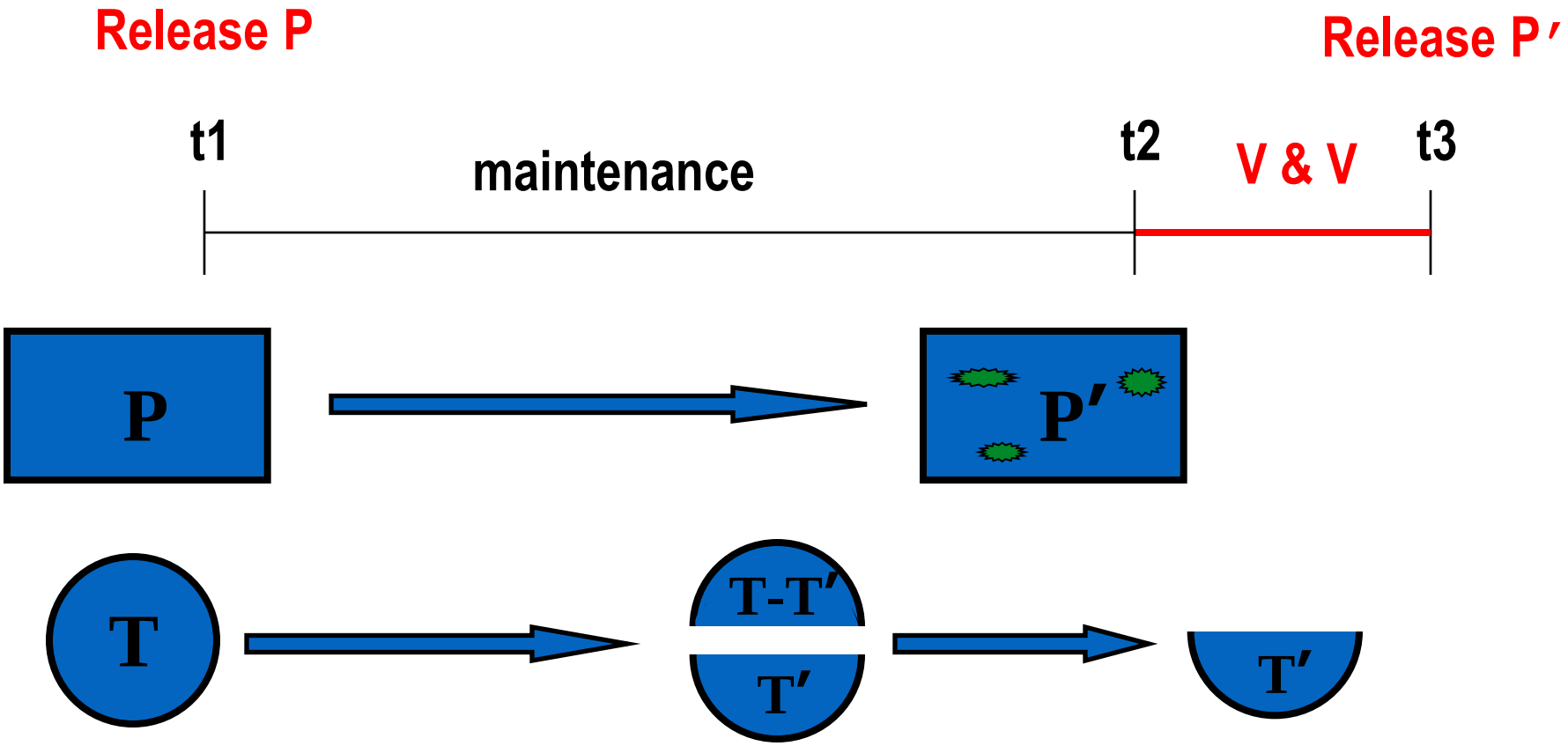
Regression Testing



From a 2009 survey:

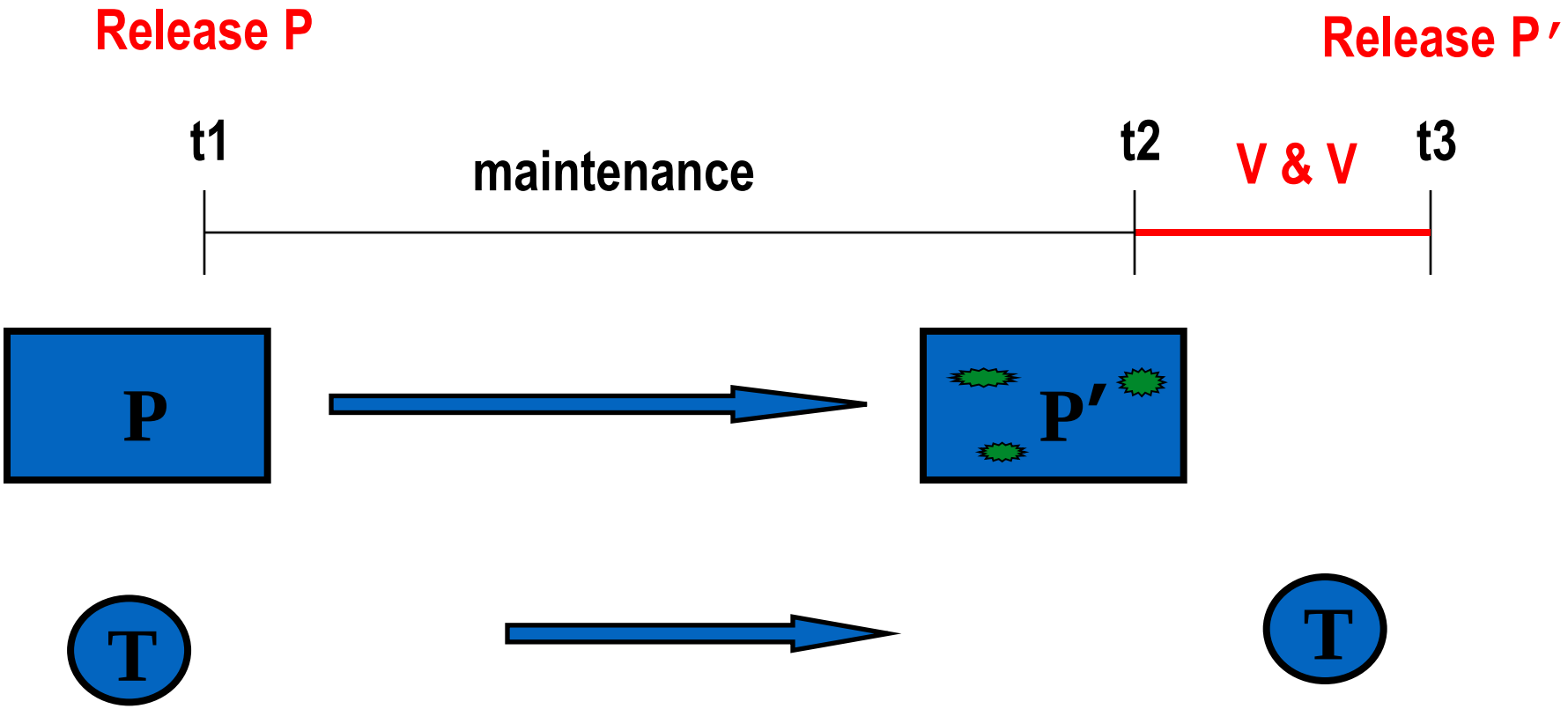
- 168 papers on regression testing

Regression Test Selection



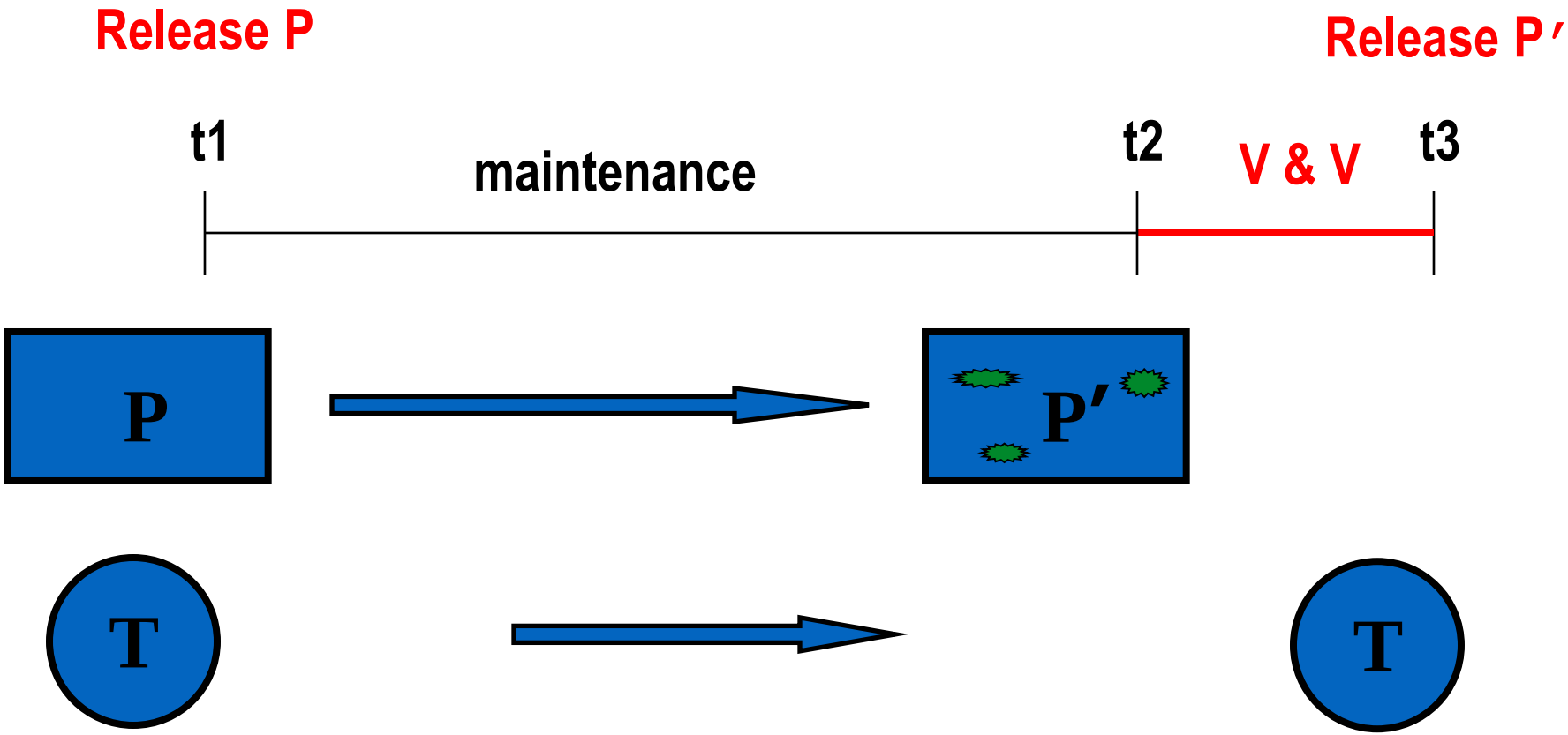
A 2009 survey: 87 papers

Test Suite Minimization



A 2009 survey: 37 papers

Test Case Prioritization



A 2009 survey: 47 papers

A 2015 Survey: over 170 papers

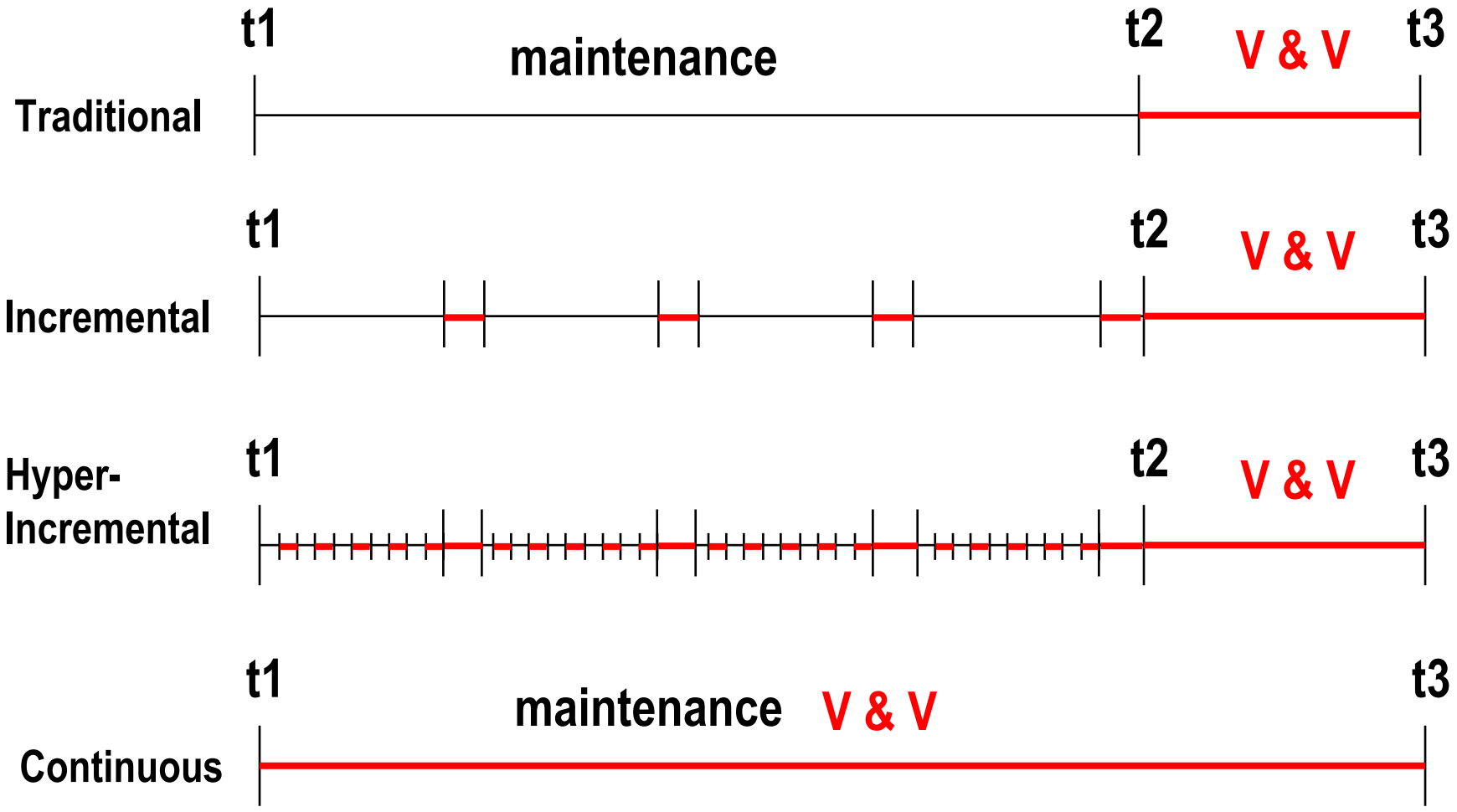
Regression Testing (What is Happening?)

- Many new techniques are using more sophisticated approaches such as search-based and AI-based
- More extensive empirical studies being conducted
- New topics:
  - Test suite augmentation
  - New software domains (e.g. web-based)

Regression Testing (What's Needed Next?)

- Greater industrial uptake
  - Requires better efforts to understand practitioners' problems and needs
  - Industrial case studies may help
- Consideration of specific practical processes

Regression Testing (Process Models)



So Many Things, So Little Time…

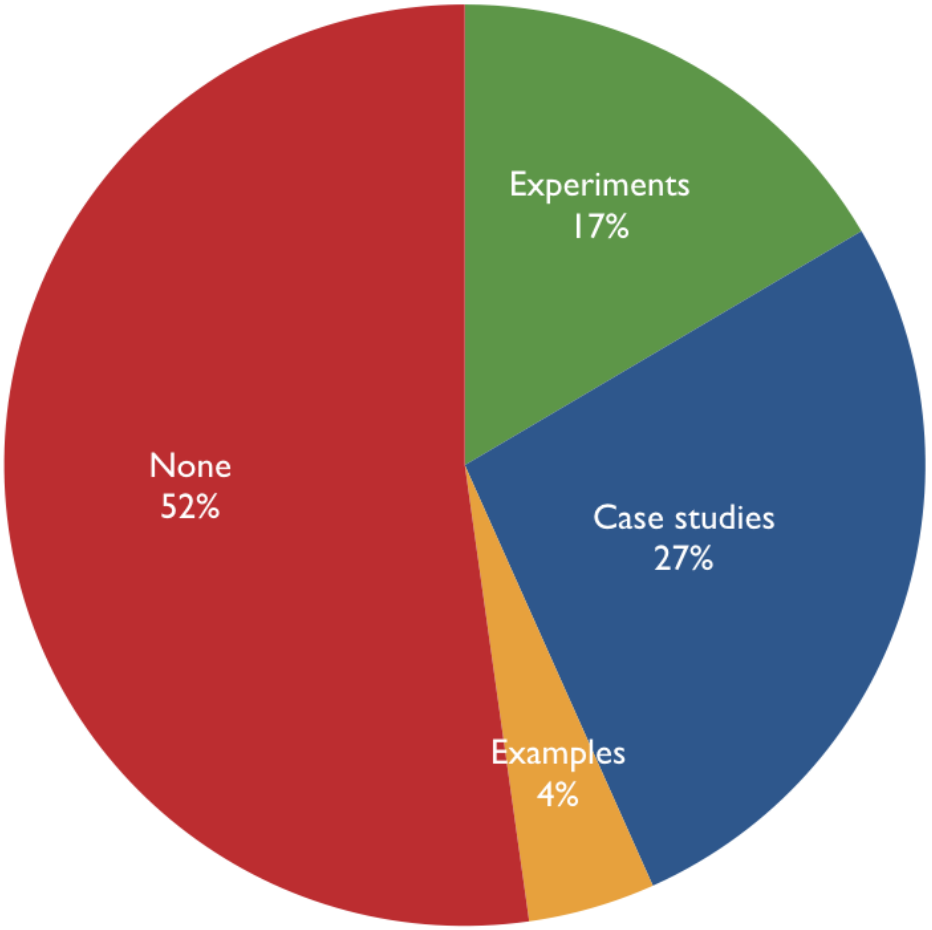
Automated Test Input Generation | Regression Testing | Empirical Studies & Infrastructure

Practical Contributions | Leveraging Cloud and Crowd



Empirical Studies & Infrastructure

- Testing is heuristic ➡ must be empirically evaluated
- State of the art in 2005: study on 224 papers on testing (1994–2003)
- Things have changed dramatically since then:
  - Empirical evaluations are almost required
  - Artifact evaluations at various conferences



Empirical Studies & Infrastructure (What Changed?)



- Increased availability of experiment objects
  - Repositories: SIR (over 2600 users registered, over 800 papers used it), Bugbench, iBugs, Marmoset, SAMATE Reference Dataset, …
  - Open-source systems, often large and available with versions, tests, bug reports, …



- Increased availability of supporting infrastructure (analysis tools, coverage tools, mutation tools, …)



- Increased understanding of empirical methodologies

Empirical Studies & Infrastructure

(What's Needed Next?)



- Address threats to external validity by doing:
  - Controlled studies of more realistic systems
  - Case studies of industrial systems



- Move beyond "tool" and "technique" studies
  - "If you build it they will come" is not necessarily the case with software engineering tools; we need to investigate whether engineers can and will actually use tools

So Many Things, So Little Time…

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Practical Contributions | Leveraging Cloud and Crowd



Practical Contributions

- Frameworks for test execution
- Shortening of the testing process life cycle

Practical Contributions

- Frameworks for test execution
  - Dramatically improved the stat of the art
  - Indirectly affected research
  - Examples:









●●●

- Shortening of the testing process life cycle

Practical Contributions

- Frameworks for test execution
- Shortening of the testing process life cycle
  - From integrating and testing “at the end”, to early integration and testing, to continuous integration (CI)
  - Widely used in industry
  - Examples:


Hudson


Jenkins



Travis

●●●

So Many Things, So Little Time…

Automated Test Input Generation | Regression Testing | Empirical Studies & Infrastructure

Practical Contributions | Leveraging Cloud and Crowd



Leveraging Cloud and Crowd



- From local to remote (data centers, servers)
- Software increasingly built and run on the net (e.g., cloud IDEs)
- Natural for testing to follow (e.g., symbolic execution, test farms, heavy-weight analysis)



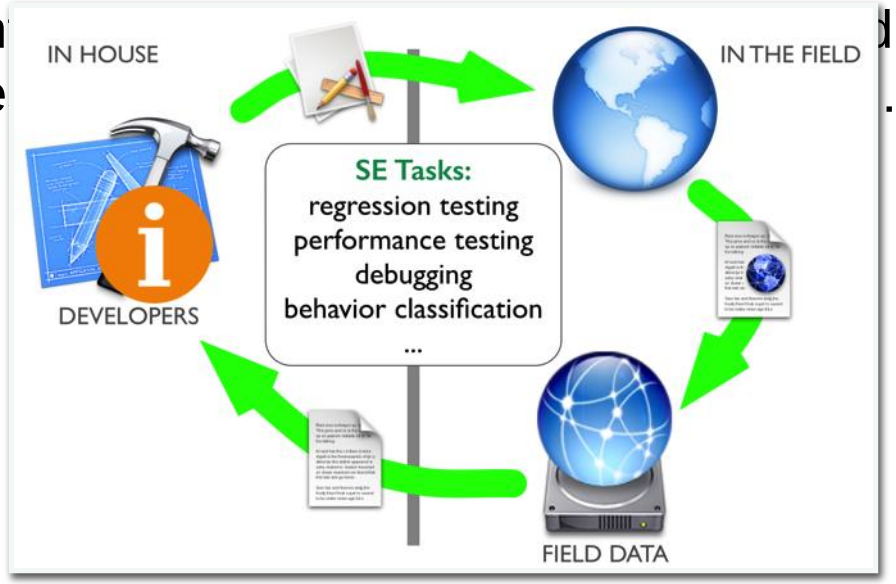




Leveraging Cloud and Crowd



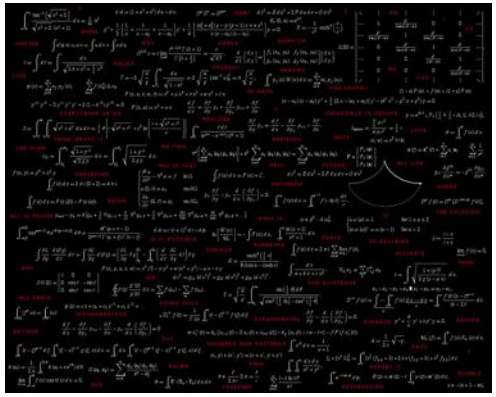
- Testing is still very much human intensive
- Leveraging the crowd makes much sense
- This has been happening for some time, both in academia and in industry
- In
  te …)



Leveraging Cloud and Crowd



- Testing is still very much human intensive
- Leveraging the crowd makes much sense
- This has been happening for some time, both in academia and in industry
- Interesting new directions (game based testing and verification, crowd oracles, …)





- Testing as a game?
- Must be a game people are willing to play
- Must be easier than the original problem

In Summary



- Incredible amount of work on testing
- Yet, things are not that different…
  …or are they?

- Automated testing
- Empirical evaluation
- Testing strategies
- Testing tools
- Testing process
- …

Future Directions



And a personal message

- Stop chasing full automation
- Testing Real-World Systems
  Oracles
- Probabilistic Approaches
  Testing Non-Functional Properties
  Domain-Based Testing
  Leveraging Cloud and Crowd
  repetitive, error-prone, etc. tasks



With much appreciated input/contributions from

- Alex Groce
- Andrea Arcuri
- Andreas Zeller
- Andy Podgurski
- Antonia Bertolino
- Atif Memon
- Corina Pasareanu
- Darko Marinov
- David Rosenblum
- Elaine Weyuker
- John Regehr
- Lionel Briand
- Lori Pollock
- Mark Grechanik
- Natalia Juristo
- Paolo Tonella
- Patrice Godefroid
- Per Runeson
- Peter Santhanam
- Phil McMinn
- Phyllis Frankl
- Robert Hierons
- Satish Chandra
- Sebastian Elbaum
- Sriram Rajamani
- T. Y. Chen
- Tom Ostrand
- Wes Masri
- Willem Visser
- Yves Le Traon

Thanks for attending this ACM SIGSOFT Webinar!

Software Testing: A Research Travelogue

- **Information on future and past SIGSOFT webinars is found at:**
  - http://www.sigsoft.org/resources/webinars.html

- Dec 17 12PM EST: *TBD*
  - Speaker: Bram Adams, Polytechnique Montreal
- Jan 20, 12PM EST: *TBD*
  - Speaker: Gail Murphy, University of British Columbia